Problem Statement:

Currently, the business relies fully on the physical store. Due to this, scalability is limited, and the order handling process is not well-structured, with manual activities being the primary method of handling orders. To expand sales channels, there is no centralized system to handle the critical business operations like order management, inventory management, order fulfilment and managing operational workflows efficiently, which will lead to potential revenue loss and high operational overhead.

Business Goals:

- Establish a single, centralized view and control of orders across all existing sales channels, with the ability to onboard additional channels without redesigning core order operations.
- Increase order fulfilment efficiency by reducing manual interventions in order validation, routing, and shipment coordination.
- Enable business growth across digital sales channels while decoupling order volume growth from linear increases in operational headcount and physical store expansion.
- Improve customer experience by reducing order delays, fulfilment errors, and post-order support interactions.

KPIs:

- Order Completion Rate (%):
    - Percentage of orders successfully processed and shipped without cancellation or manual rework, measured from order creation to shipment confirmation.
    - **Target:** ≥ 95%

- Order-to-Shipment Cycle Time:
    - Average time elapsed between order creation and shipment confirmation for successfully fulfilled orders.
    - **Target:** Reduce from X to Y

- Manual Intervention Rate (%)
    - Percentage of orders requiring at least one manual operational action outside the standard automated workflow.
    - **Target:** Reduce by X% from baseline.

Business Requirements:

- The business must be able to validate, allocate, and update order status for all orders through a centralized operational workflow without reliance on external spreadsheets.
- The business must be able to define and execute automated order routing and fulfilment decision rules based on configurable business criteria.

- The business must be able to onboard and operate additional online sales channels while maintaining existing order processing workflows and operational controls.
- The business must be able to identify, monitor, and manage orders that are at risk of breaching defined fulfilment SLAs prior to customer impact.

Stakeholder List:

Primary Business Stakeholders:

- **Business Owner**
  Owns overall business objectives, revenue targets, and success metrics of the OMS initiative.

- **Product Owner**
  Defines product scope, prioritizes requirements, and aligns OMS capabilities with business roadmap.

Operational Stakeholders:

- **Operations Team**
  Manages order processing, exception handling, manual interventions, and day-to-day OMS operations.

- **Customer Support Team**
  Handles customer inquiries related to order status, delivery delays, payment failures, and returns.

- **Finance Team**
  Responsible for payment reconciliation, refunds, settlements, and financial reporting related to orders.

External Business Partners:

- **Logistics / Courier Vendors**
  Responsible for shipment execution, delivery updates, and return (RTO) handling.

- **Warehouse Partners**
  Manage inventory, picking, packing, and dispatch activities.

- **Payment Service Providers**
  Enable online payment processing and transaction settlements.

In-scope:

- Ingestion and centralization of order data from supported sales channels into the Order Management System (OMS)
- Visibility of inventory availability within the OMS through synchronization with upstream or downstream inventory systems
- Centralized operational management of orders, including validation, status updates, and exception handling
- Exchange of order information with warehouse and logistics systems to support fulfillment activities
- Propagation of order status updates to customer-facing channels where supported
- Definition of system workflows and controls to reduce manual order handling during operations

Out-of-scope:

- End-to-end automation of all operational workflows without manual intervention
- Returns, refunds, and reverse logistics processes
- Enhancements or feature additions beyond the defined Phase 1 requirements after scope sign-off

Assumptions:

Serial No	Assumptions	Reason	Risk if Wrong
1	Customers have access to internet-enabled devices (web/mobile)	Online OMS success depends on digital adoption	Low adoption, which may lead to revenue target miss
2	Customers will accept guest checkout in Phase 1	Chosen to reduce friction and launch faster	High guest usage, which will result in limited customer insights
3	ERP, warehouse, and logistics systems expose usable APIs	OMS depends on system-to-system sync	Manual work increases, sync delays
4	Payment gateway supports required payment methods and SLAs	Order success rate depends on payment reliability	Order failures, revenue loss
5	Operations and warehouse teams follow defined manual fallback processes	Phase 1 includes partial automation	Process breakdowns, order delays
6	Product, inventory, and pricing	OMS reflects	Customer complaints,

	data are reasonably accurate at source	upstream data	order errors
7	Business teams will own data correctness	OMS will not correct upstream master data in Phase 1	Incorrect data may result in wrong orders, customer dissatisfaction, increased operational rework, and revenue loss

Dependencies:

Serial No.	Dependency	Dependency Type	Impact on OMS
1	Payment Gateway availability and SLA	External / Technical	Blocks order placement
2	ERP system API readiness	External / Technical	Blocks order & inventory sync
3	Warehouse system integration	External / Ops	Impacts inventory accuracy
4	Logistics partner tracking updates	External / Vendor	Impacts delivery visibility
5	SMS / Email notification service	External / Vendor	Affects customer communication
6	Dev team availability for Phase 1	Internal / Resource	Delays delivery
7	QA & Security validation	Internal / Compliance	Go-live is blocked if it fails

Constraints:

1. Time Constraints:
    a. Phase 1 must launch by **20 December 2025** to support business expansion.
    b. No extension beyond peak sales periods (sale events, launches).

2. Budget Constraints:
    a. Limited Phase 1 budget restrictions:
        i. Full automation

    ii. Real-time integrations
    iii. Additional infrastructure

3. Technical Constraints:
    a. PCI compliance is mandatory
    b. Batch sync interval limited to **15 minutes** due to Operations SLA.
    c. The system must handle **5000 traffic/hour** without degradation.

4. Operational Constraints:
    a. Manual fallback is limited to trained staff only.
    b. Guest checkout is mandatory in Phase 1.
    c. Customer account management deferred to Phase 2.

Non-functional requirements:

- The solution must comply with applicable PCI-DSS requirements for handling payment-related data.
- The solution must implement controls to prevent unauthorized access, automated order abuse, and malicious request patterns.
- The solution must support at least 5,000 concurrent order-related requests per hour without degradation of defined performance SLAs.
- The solution must process standard order operations within acceptable response times under peak load conditions.

Risks:

1. External System & Vendor Dependencies:
   The biggest risk is dependency on external systems like ERP, payment gateways, and logistics partners. If their APIs are unstable, unavailable, or change unexpectedly, core OMS flows, like order sync and delivery updates, will fail.

   Mitigation:
    a. Batch sync instead of real-time in Phase 1
    b. Manual fallback for critical operations
    c. Clear SLAs with vendors

2. Data Quality & Sync Reliability:
   OMS heavily depends on accurate inventory, pricing, and order data. If upstream data is inaccurate or delayed, it can cause order failures, stock mismatches, and customer dissatisfaction.

   Mitigation:
    a. Defined retry mechanisms
    b. Error notifications to the Operations team.

   c. Operational validation checks

3. Operational Readiness & Adoption:
   Even with a working system, lack of training or unclear manual fallback processes can derail day-to-day operations, especially in Phase 1, where automation is partial.

   Mitigation:
   a. Documented fallback processes
   b. Limited automation in Phase 1
   c. Operations team involvement early

4. Scope Creep & Unrealistic Expectations:
   Uncontrolled scope changes or pressure for full automation in Phase 1 could delay launch and increase cost without immediate business value.

   Mitigation:
   a. Clear Phase 1 vs Phase 2 scope
   b. Explicit trade-offs
   c. KPI-driven prioritization

Trade-offs Summary:

1. Guest checkout vs login mandatory
   a. Trade-off:
      Launch with guest checkout for the customer over the mandatory login feature
      i. Option A:
         Website and mobile app launch with a guest checkout feature for users
      ii. Option B:
         Website and mobile app launch with a mandatory login feature for users to place orders
      iii. Option C:
         Website and mobile app launch with both guest and basic login features (optional)

   b. Trade-off analysis:
      i. Option A helps launch the system faster in the market, allowing users to place orders quickly as guests, at the cost of limited customer account access in the initial phase.
      ii. Option B allows the system to track all customers' activity on the website and mobile app, as login is mandatory at the cost of drop-off, because not all customers may want to proceed as logged in.
      iii. Option C allows users to place orders both as guests and logged in, and it also helps the business to get more users and start generating revenue at the cost of timeline to launch the system.

c. Decision Impact:
Option A is preferred as the launch of the first phase to start early business revenue, but with the risk of not having any customer account access to validate the order information in the portal itself.

2. Real-time sync vs near-real-time batch sync
    a. Trade-off:
    Near-real-time batch sync in place of real-time sync
        i. Option A:
        Order information syncs with external systems via a batch process at a regular interval.
        ii. Option B:
        Order information syncs with the external systems in real-time immediately after order placement.

    b. Trade-off analysis:
        i. Option A should support reducing the number of API calls, enabling retry logic in case an error happens, and improving operational stability at the cost of delayed data transfer to the external systems.
        ii. Option B enables the real-time data sync to the external systems without any delay at the cost of higher system maintenance costs and higher system load due to an increased number of API calls.

    c. Decision Impact:
    Option A is suggested as the ideal data sync process among the multiple systems to improve system performance and prevent higher system maintenance costs, but with the risk of delayed data transfer to the other systems.

3. Manual fallback vs full automation in Phase 1
    a. Trade-off:
    Manual fallback in place of full automation in the first phase
        i. Option A:
        Manual interventions are required to continue the business operations when the automation system faces sudden challenges.
        ii. Option B:
        Fully automated system to support business operations with no planned manual fallback.

    b. Trade-off analysis:
        i. Option A supports the business teams to continue the critical operations in case issues occur with the automation process, but at the cost of higher operational efforts and future enhancements to upgrade the automation system.

ii. Option B supports the system to operate all the required business activities without any manual interventions, but at the cost of a higher implementation timeline and testing efforts, delayed system launch and higher development and maintenance time and cost.

c. Decision Impact:
Option A is preferred over Option B in phase 1 to expand the business with the online portals faster and generate revenue, but with the risk of increased operational activities with limited scalability, with the scope of future enhancements to improve the automation system.

Success Metrics:

1. An increase in successfully placed orders by 10 - 15% based on the current baseline of 60%
2. Reduce the percentage of orders requiring manual operational intervention by 40–50% compared to the pre-implementation baseline.
3. Reduce the average time taken to synchronize order data from order creation to availability in downstream warehouse and logistics systems from ~25 minutes to 10–15 minutes.
4. Reduce the average time required to complete order processing activities (from order creation to fulfilment handoff) from ~30 minutes to ~15 minutes.
5. Improvement of orders sync success rate to 90 - 95% across all the integrated systems

All success metrics are indicative and will be validated against actual baseline data during discovery and post-deployment monitoring.